Meteorology: Storm & Rainfall Pattern Analysis

A SQL- and R-based analysis of a decade of Australian daily weather observations, identifying seasonal rainfall patterns, geographic hotspots, and the atmospheric signature of extreme rainfall events.

Project: Data Analytics Week 3 — Advanced Excel, SQL & R
Dataset: weatherAUS.csv — 145,460 daily observations, 49 Australian stations, Nov 2007–Jun 2017
Tools: PostgreSQL 18, R (RPostgres, dplyr, ggplot2)

1. Objective

This project analyses approximately ten years of daily weather observations from 49 Australian weather stations to characterise rainfall behaviour at a national level. The analysis was scoped around four guiding questions:

- Which locations receive the most rainfall, and is that rainfall frequent or intense?
- Is there a consistent seasonal (monthly) rainfall pattern across Australia?
- Has average rainfall trended upward, downward, or stayed flat over 2009–2016?
- What defines an “extreme” rainfall day, where do such days cluster, and what atmospheric conditions accompany them?

The intended audience is a general policy/public-reporting context, consistent with the brief for Project 10 – *Meteorology: Storm/Rainfall Pattern Analysis*.

2. Methodology

The workflow followed a three-stage pipeline: SQL for storage, cleaning and aggregation; R for statistical testing and visualisation; Excel for dashboarding (Phase 4, separate deliverable).

2.1 Data Source

The **weatherAUS** dataset (Kaggle) was imported into a PostgreSQL database (`weather_aus` table, 145,460 rows, 23 columns). The table spans 2007-11-01 to 2017-06-25 across 49 distinct weather stations.

2.2 Data Quality Finding

An initial NULL-count audit suggested the data was almost entirely complete – an unexpected result given Kaggle's own documentation, which states that columns such as *Sunshine* and *Evaporation* are roughly 40–48% missing in the source CSV. Further investigation revealed the cause: during import, missing values (originally the string “NA”) had been coerced into numeric columns as the literal floating-point value `NaN` rather than a true SQL `NULL`. Because SQL's `COUNT()` only skips true NULLs, this silently hid the missing data from a standard completeness check, and would have caused every `AVG()`/`SUM()` touching an affected row to silently return `NaN`.

A targeted check confirmed the scale of the issue:

Column	NaN count	% of rows
sunshine_hrs	69,835	48.0%
cloud_3pm	59,358	40.8%
cloud_9am	55,888	38.4%
evaporation	62,790	43.2%
pressure_9am / pressure_3pm	15,065 / 15,028	10.4% / 10.3%
wind_gust_spd	10,263	7.1%
rainfall_mm	3,261	2.2%

Resolution: a corrected view, `weather_aus_clean`, was created in PostgreSQL converting every affected `NaN` to a true `NULL` via conditional `CASE` expressions. All subsequent queries and exports reference this cleaned view, not the raw table.

2.3 Integrity Checks

- **Duplicate records:** no location/date pairs appeared more than once – confirmed via a GROUP BY/HAVING check.
- **Categorical integrity:** `rain_today` / `rain_tomorrow` contained only clean “Yes”/“No” values, with missing values matching exactly the rows missing `rainfall_mm` (3,261 rows in both cases) – confirming these flags are derived directly from rainfall and behave consistently.

2.4 Extreme Event Definition

Two complementary definitions of an “extreme rainfall day” were used and cross-checked against each other:

Approach	Threshold	Days meeting it	% of rainy obs.
Data-driven (99th percentile)	≥ 37.4 mm	1,435	1.0%
BOM “heavy rain”	≥ 25 mm	2,904	2.0%
BOM “very heavy rain”	≥ 50 mm	808	0.57%
BOM “extreme/torrential”	≥ 100 mm	151	0.11%

The 99th-percentile cutoff (37.4 mm) sits between the Bureau of Meteorology's “heavy” (25 mm) and “very heavy” (50 mm) classifications, while the 99.9th percentile (102 mm) closely matches the BOM “extreme” threshold of 100 mm. The two methods are therefore broadly consistent, with the BOM scale offering the more interpretable, report-ready tiers used throughout the remainder of this analysis.

3. Key Findings

151	49	Cairns	Feb
Extreme days (≥100mm) nationally, 2007–2017	Weather stations analysed	Wettest location (5.74mm/day avg)	Peak rainfall month nationally

3.1 Geographic Pattern: Where It Rains Most

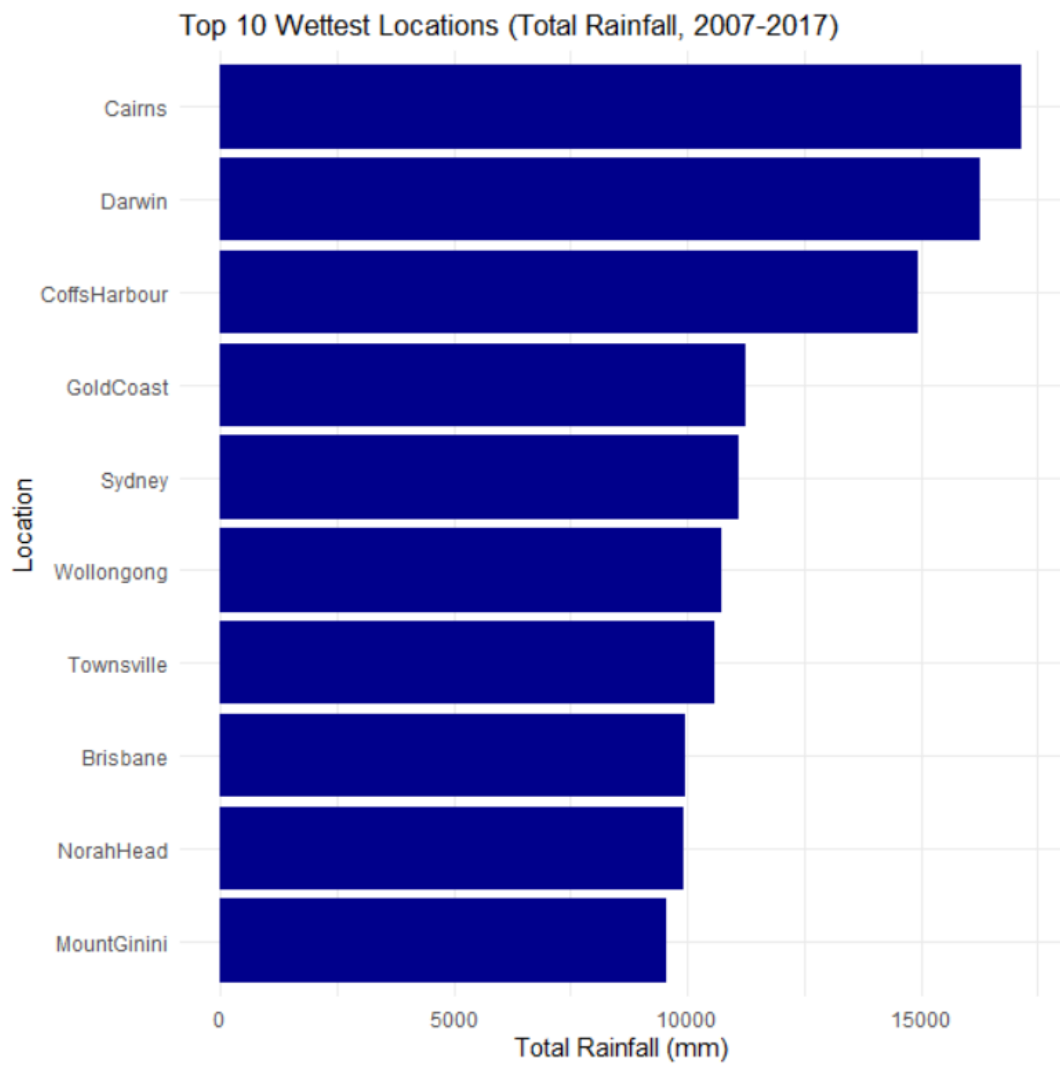

Figure 1. Top 10 wettest locations by total rainfall, 2007–2017.

Rainfall totals are dominated by tropical northern stations (Cairns, Darwin) and the eastern seaboard (Coffs Harbour, Gold Coast, Sydney, Wollongong, Townsville, Brisbane) – the locations most exposed to monsoon troughs, tropical lows, and east-coast low-pressure systems. No southern or inland temperate station appears in the top 10, indicating that Australia's heaviest rainfall is structurally a tropical/coastal phenomenon rather than a nationally uniform one.

3.2 Seasonal Pattern: When It Rains Most

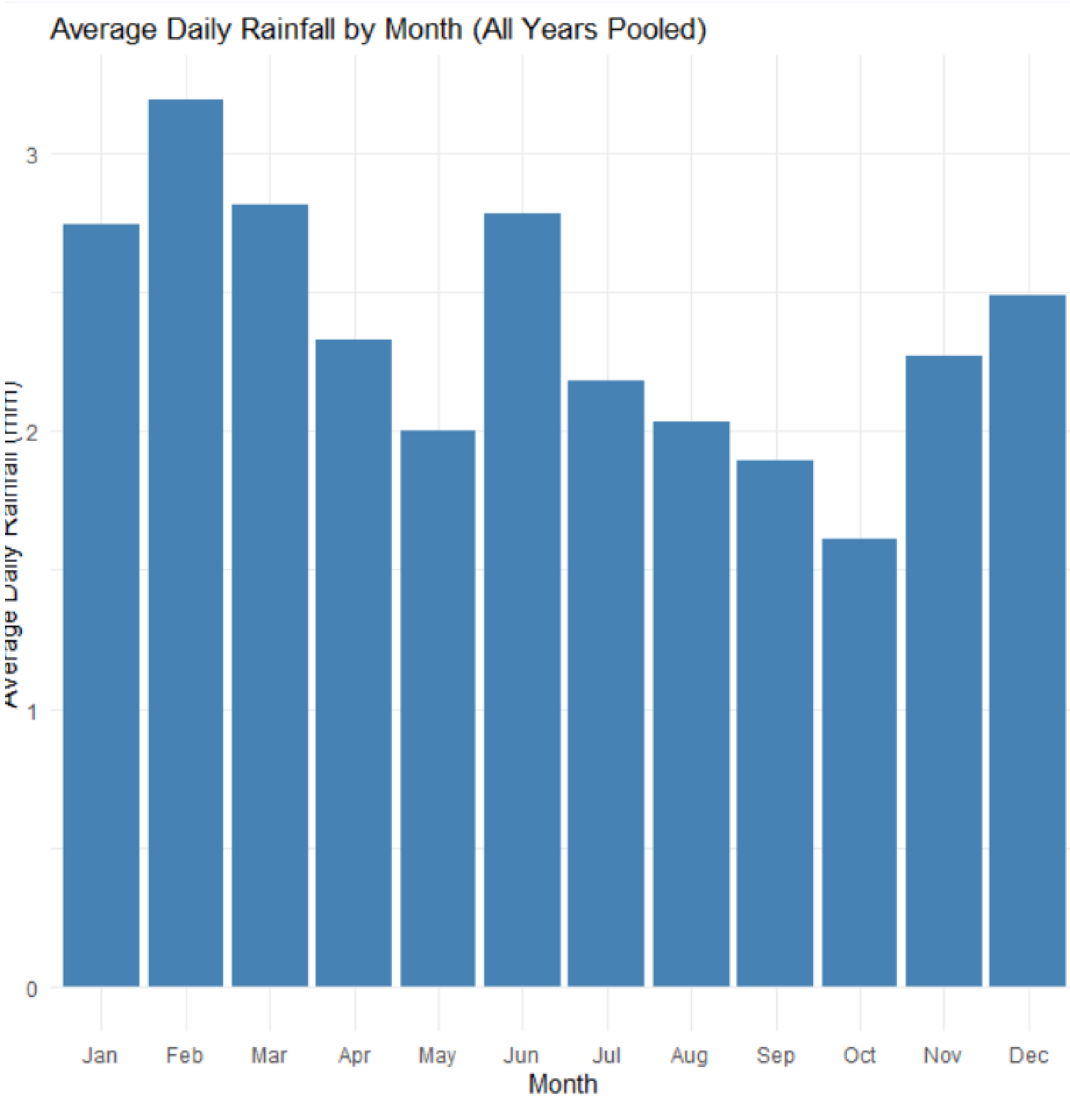

Figure 2. Average daily rainfall by month, all years and stations pooled.

February is the wettest month nationally (3.19 mm/day average), consistent with the peak of the northern Australian monsoon and cyclone season. October is the driest (1.61 mm/day), sitting in the seasonal gap between southern winter rains ending and the northern wet season not yet beginning. A secondary, smaller peak appears in June (2.78 mm/day) – this reflects a second, independent rainfall mechanism: frontal winter rain in southern temperate regions (e.g. Melbourne, Adelaide), which is more frequent but less intense than tropical downpours. Notably, June recorded more rainy days (3,267) than February (2,176) despite lower average intensity, illustrating a frequency-versus-intensity contrast between the two rainfall regimes.

3.3 Yearly Trend: Is Rainfall Changing Over Time?

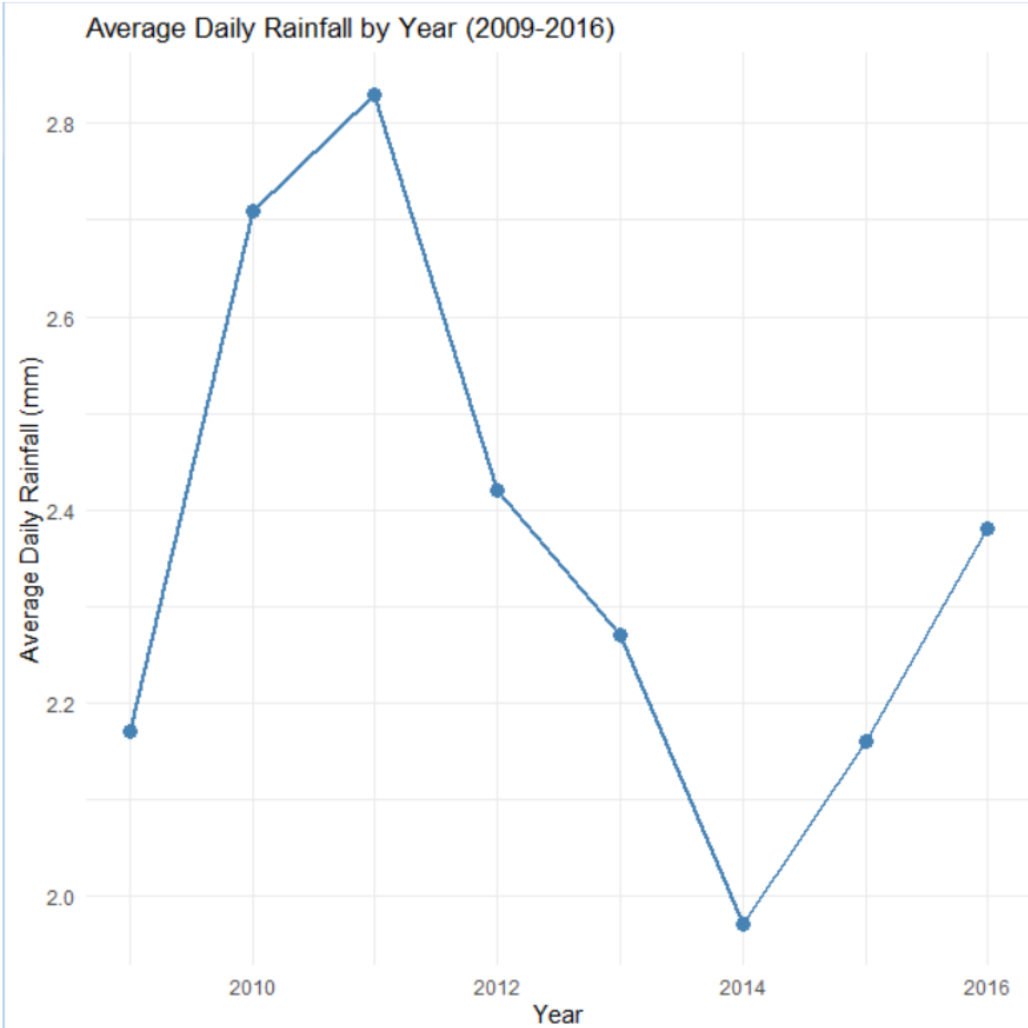

Figure 3. Average daily rainfall by year, 2009–2016 (restricted to years with stable station coverage).

Average daily rainfall shows no steady long-term trend: it rises from 2009 to a peak in 2011 (2.83 mm/day), falls to a trough in 2014 (1.97 mm/day), then partially recovers by 2016. A linear regression of average rainfall against year confirmed this visual impression statistically:

Statistic	Value	Interpretation
Slope (mm/year)	–0.048	Slight apparent decline, but…
p-value	0.32	Not statistically significant (> 0.05)
R^2	0.16	Only 16% of variation explained by year

Conclusion: there is no statistically detectable directional trend in this period. The peak year, 2011, coincides with the well-documented 2010–2011 La Niña event and associated eastern-Australia flooding, while 2014 (the driest year) is consistent with neutral/El Niño-leaning conditions. This suggests year-to-year rainfall variability in this dataset is driven primarily by short climate cycles (El Niño–La Niña) rather than a longer-term trend. With only eight annual data points, however, this test has limited statistical power, and the result should be read as “no trend detected” rather than “no trend exists.”

3.4 Extreme Events: Where, When, and Under What Conditions

Of 142,199 valid daily rainfall observations, 151 (0.11%) reached the BOM "extreme/torrential" threshold of 100 mm or more. These events were heavily concentrated by both location and season:

Location	Month	Extreme days (≥100mm)
Darwin	February	8
Cairns	February	7
Cairns	January	7
Cairns	March	5
Coffs Harbour	Jan / Feb / Mar	4 each
Brisbane	January	4
Darwin	January	4
Gold Coast	April	4
Townsville	Feb / Mar	4 each

Every entry in the top of this ranking is a tropical or sub-tropical station, and every one falls in the January–April window – the Australian monsoon and tropical cyclone season. No temperate or southern station appears, reinforcing the finding from Section 3.1 that extreme single-day rainfall in Australia is a tropical/sub-tropical, wet-season phenomenon.

To test the physical plausibility of this pattern, same-day morning pressure and humidity were correlated with rainfall amount across all heavy-rain days (≥25 mm, n=2,904):

Variable	Correlation with rainfall	p-value	Direction
Humidity (9am)	r = 0.117	< 0.001	Higher humidity → heavier rain
Pressure (9am)	r = –0.107	< 0.001	Lower pressure → heavier rain

Both correlations are statistically significant but individually modest in size. This is expected: the test is restricted to days that are already heavy-rain days, where humidity is typically already elevated and pressure already suppressed – reducing the remaining variation available to explain differences in rainfall amount. Humidity and pressure likely matter more for predicting *whether* a heavy rain day occurs at all than for predicting *how heavy* it becomes once one is underway. Nonetheless, the direction of both effects is exactly as expected for monsoon troughs and tropical low-pressure systems, providing a physically consistent explanation that ties together the geographic, seasonal, and atmospheric findings.

4. Interpretation & Synthesis

Three independent lines of evidence in this analysis point to the same underlying mechanism for Australia's most significant rainfall:

- **Geography:** the wettest locations and the locations with the most extreme rainfall days are tropical and sub-tropical (Cairns, Darwin, Townsville, Coffs Harbour) – not temperate or inland.
- **Timing:** both average rainfall and extreme-event frequency peak in the January–March window, aligning with the Australian monsoon and cyclone season.
- **Atmospheric mechanism:** heavier rainfall days are measurably associated with lower pressure and higher humidity – the textbook signature of monsoon troughs and tropical low-pressure systems.

Year-to-year variation, by contrast, does not show a directional trend over the available window; the wettest (2011) and driest (2014) years align with known El Niño–La Niña phases rather than a steady climate drift. This distinction matters for how the results should be communicated: Australia's rainfall extremes are reliably predictable by *location and season*, but not by a simple long-term trajectory over this ten-year window.

4.1 Limitations

- The yearly trend test (n=8) has limited statistical power; a real but modest trend could exist without reaching significance.
- 2007, 2008, and 2017 were excluded from the yearly trend due to inconsistent station coverage in those years, slightly shortening the usable time series.
- Correlation analysis used same-day 9am readings only; lagged or 3pm conditions, and wind direction/speed, were not modelled and may add explanatory power.
- One extreme outlier (a single-day evaporation reading of 145 mm) was identified during data quality review but not yet investigated – it does not affect the rainfall analysis but should be reviewed before evaporation data is used in future work.

4.2 Recommended Next Steps

- Build the Excel dashboard (Phase 4) translating Figures 1–3 into an interactive PivotTable/slicer-driven view for non-technical stakeholders.
- Extend the atmospheric correlation analysis to include wind gust speed/direction and 3pm readings, and test lagged relationships (e.g. does a pressure drop the day before predict next-day extreme rainfall, leveraging the existing RainTomorrow field).
- Investigate the 145mm evaporation outlier and any other extreme outliers before using evaporation or sunshine data in further modelling.

Prepared as part of the Data Analytics Week 3 Project (Advanced Excel, SQL & R) – Project 10: Meteorology, Storm/Rainfall Pattern Analysis. Source data: weatherAUS.csv (Kaggle).